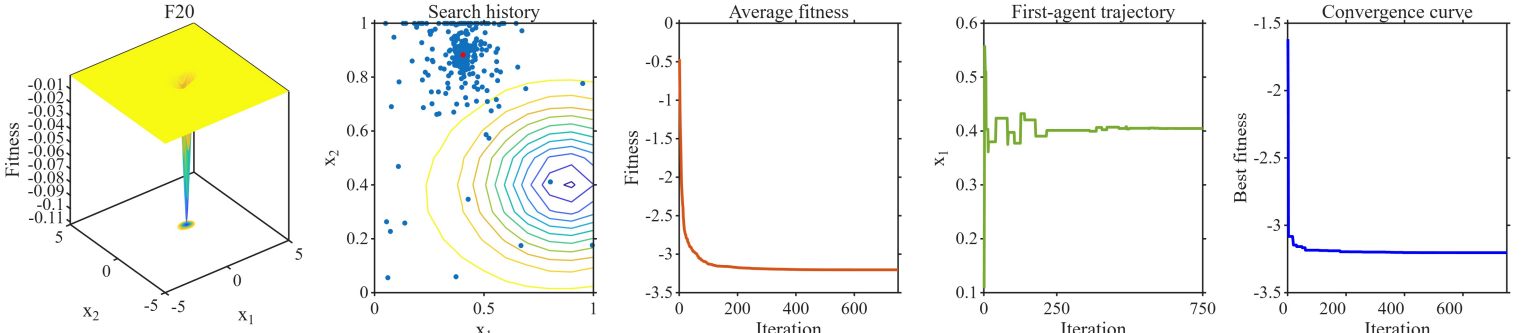

F20
Fitness
-0.01
-0.02
-0.03
-0.04
-0.05
-0.06
-0.07
-0.08
-0.09
-0.1
-0.11
5
0
-5
x2
-5
0
5
x1
Search history
x2
1
0.8
0.6
0.4
0.2
0
0
0.5
1
x1
Average fitness
Fitness
0
-0.5
-1
-1.5
-2
-2.5
-3
-3.5
0
200
400
600
Iteration
First-agent trajectory
x1
0.6
0.5
0.4
0.3
0.2
0.1
0
250
750
Iteration
Convergence curve
Best fitness
-1.5
-2
-2.5
-3
-3.5
0
200
400
600
Iteration